26900347 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCMAT301 - LIMITS AND CONVERGENCE

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **5** questions

Each question carries **2** marks

1. Find $\lim_{n\longrightarrow\infty}\left(\frac{2n+1}{n}\right)$. [A] / [CO1]

2. Write the tail of the sequence $(\frac{1}{n})$ beginning with x_5 [U] / [CO1]

3. Why is the boundedness condition essential in the Bolzano–Weierstrass Theorem? [U] / [CO3]

4. State the theorem on the proper divergence of monotone sequences. What can be concluded if a monotone sequence is unbounded above or unbounded below? [U] / [CO5]

5. Prove that the ratio test fails to check the convergence of the p - series. [An] / [CO6]

6. Discuss the conditions needed for Abel's Test. [U] / [CO7]

7. Explain the meaning of $\lim_{x\to-\infty} f(x) = L.$ [U] / [CO8]

8. What is the limit of the function $f(x) = \frac{sinx}{x}$ as $x \to 0$.? [U] / [CO8]

[2x5 = 10]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **6** marks

9. Prove that every convergent sequence of real numbers is bounded. [A] / [CO1]

10. Prove that, if (x_n) is a bounded increasing sequence, then $\lim(x_n) = \sup\{x_n : n \in \mathbb{N}\}$. [A] / [CO2]

11. Let (x_n) and (y_n) be two sequences of positive real numbers and suppose that for some $L \in \mathbb{R}, L \geq 0, \lim\left(\frac{x_n}{y_n}\right) = L$. Using the definition of a properly divergent sequence, show that if $\lim(y_n) = +\infty$ then $\lim(x_n) = +\infty$. [A] / [CO5]

12. (a) Define a Cauchy's sequence [A] / [CO4]
(b) If $x_1 = 1,\ x_2 = 2$ and $x_n = \frac{1}{2}(x_{n-2} - x_{n-1})$ for $n > 2,$ prove that $|x_n - x_{n+1}| = \frac{1}{2^{n-1}}$ for $n \in \mathbb{R}$.

13. Prove that absolute convergence of a series implies convergence, but not conversely. [An] / [CO6]

14. Use Raabe's test to determine whether the series $\sum_{n=1}^{\infty} \frac{1}{n(n+3)}$ converges or diverges. [U] / [CO6]

15. Prove squeeze theorem for functions. [K] / [CO8]

16. Explain the limit theorem for quotient of two functions and explain why it cannot be directly use to find the limit of $\frac{x^2-4}{3x-6}$ at x=2. Also find the limit. [U] / [CO8]

[6x5 = 30]

Part C

Essay Type Questions

Answer any **3** questions

Each question carries **10** marks

17. Use the definition of the limit of a sequence to prove the following limits [A] / [CO1]
a) $lim(\sqrt{n+1} - \sqrt{n}) = 0$
b) $lim(\frac{3n+1}{n+1}) = 3$

18. (a) Prove that every Cauchy sequence of real numbers is bounded. Is the converse true? [An] / [CO4]
(b) State and prove Cauchy criterion for convergence of a sequence.

19. Let $X = (x_n)$ be a sequence of real numbers. Then prove that the following statements are equivallent: [An] / [CO3]

(i) The sequence $X = (x_n)$ does not converge to $x \in \mathbb{R}$.

(ii) There exists an $\epsilon_0 > 0$ such that for any $k \in \mathbb{N}$, there exists $n_k \in \mathbb{N}$ such that $n_k \geq k$ and $|x_{n_k} - x| \geq \epsilon_0$.

(iii) There exists an $\epsilon_0 > 0$ and a subsequence $X' = (x_{n_k})$ of X such that $|x_{n_k} - x| \geq \epsilon_0$ for all $k \in \mathbb{N}$.

20. Discuss the convergence and divergence of the geometric series . [A] / [CO5]

21. (a) Describe the right-hand and left-hand limits of $f(x) = \operatorname{sgn}(x)$ at $x = 0$ and state whether they are distinct. [U] / [CO8]

(b) Describe the behavior of the one-sided limits of $g(x) = e^{1/x}$ at $x = 0$.

22. Let $A \subset R, f : A \to R$ and let c be a cluster point of A . Then prove that the following statements are equivalent. [U] / [CO8]

(i) $\lim_{x \to c} f(x) = L$.

(ii) For every sequence (x_n) in A that converges to c such that $x_n \neq c$ for all $n \in N$, the sequence $(f(x_n))$ converges to L .

[10x3 = 30]